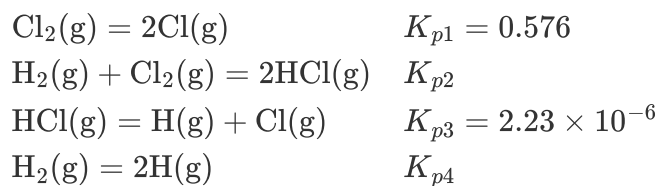


Question

Place 0.100 mol H_2 and 0.100 mol Cl_2 in a pre-evacuated special 5 L closed container, heat to 2000 K, the system reaches equilibrium, and the total pressure is 6.70 bar (1 bar = 100 kPa). The following reaction relationships exist in the system:



Which of the following statements is incorrect:

1. The total pressure of the system at 2000 K when $\text{H}_2(\text{g})$ and $\text{Cl}_2(\text{g})$ are not decomposed is 6.65 bar.
2. The equilibrium partial pressure of $\text{Cl}_2(\text{g})$ is 0.0651 bar.
3. There are 3 independent reactions in the system.
4. $K_{p4} = 4.08 \times 10^{-7}$.

A. 1

B. 2

C. 3

D. 4

E. 1, 2

F. 1, 3

G. 1, 4

H. 2, 3

I. 2, 4

J. 3, 4

K. 1, 2, 3

L. 1, 3, 4

M. 2, 3, 4

N. 1, 2, 4

O. 1, 2, 3, 4

P. No incorrect options.

Answer

Correct Answer: **I**

Detailed Explanation

To simplify the processing, the following symbols are used in the calculations, and in the equilibrium expression, all partial pressure terms are divided by the standard partial pressure by default:

$\text{Cl}_2(\text{g})$ initial partial pressure x_0 ; $\text{Cl}_2(\text{g})$ equilibrium partial pressure x_1 ; $\text{Cl}(\text{g})$ equilibrium partial pressure x_2 ;

$\text{H}_2(\text{g})$ initial partial pressure y_0 ; $\text{H}_2(\text{g})$ equilibrium partial pressure y_1 ; $\text{H}(\text{g})$ equilibrium partial pressure y_2 ;

$\text{HCl}(\text{g})$ equilibrium partial pressure z ;

Total system pressure p_t ;

From the ideal gas equation of state, we can obtain:

$$p_0 = \frac{n_0 RT}{V} = 0.2 \times 8.314 \times 2000 \div 5 \times 10^3 \text{ Pa} = 6.65 \text{ bar}$$

CHECKPOINT

1 PTS

$$p_0 = 6.65 \text{ bar}$$

1 Correct.

From the reaction equilibrium, we know that

$$x_2^2 = 0.576 x_1 \cdots (1),$$

$$x_2 y_2 = 2.23 \times 10^{-6} z \cdots (2)$$

CHECKPOINT

1 PTS

$$x_2^2 = 0.576x_1, x_2y_2 = 2.23 \times 10^{-6}z$$

From the conservation of mass, we know that

$$2x_1 + x_2 = 2y_1 + y_2 \cdots (3)$$

From the change in pressure, we know that

$$p_t - p_0 = 0.5x_2 + 0.5y_2 \cdots (4)$$

From the total system pressure, we know that

$$x_1 + x_2 + y_1 + y_2 + z = p_t \cdots (5)$$

From (1), we know that $x_1 = x_2^2/0.576 \cdots (6)$

From (4), we know that $y_2 = 2(p_t - p_0) - x_2 \cdots (7)$

Substituting (7) into (2), we know that $z = \frac{x_2[2(p_t - p_0) - x_2]}{(2.23 \times 10^{-6})} \cdots (8)$

CHECKPOINT

1 PTS

$$z = \frac{x_2[2(p_t - p_0) - x_2]}{(2.23 \times 10^{-6})}$$

Substituting (6) and (7) into (3), we know that $y_1 = x_2^2/0.576 + x_2 - (p_t - p_0) \cdots (9)$

CHECKPOINT

1 PTS

$$y_1 = x_2^2/0.576 + x_2 - (p_t - p_0)$$

Substituting (6)(7)(8)(9) into (5) yields the equation:

$$\frac{2x_2^2}{0.576} + (p_t - p_0) + \frac{x_2[2(p_t - p_0) - x_2]}{(2.23 \times 10^{-6})} = p_t$$

Substituting the data, we get: $x_2 = 0.0998$ bar

CHECKPOINT

1 PTS

$$x_2 = 0.0998 \text{ bar}$$

Then $x_1 = 0.0173$ bar, $y_1 = 0.0672$ bar, , $y_2 = 1.48 \times 10^{-4}$ bar, $z = 6.5$ bar .

CHECKPOINT

1 PTS

The equilibrium partial pressure of $\text{Cl}_2(\text{g})$ is 0.0173 bar .

2 Incorrect.

$$K_{p2} = \frac{z^2}{x_1 y_1} = 3.63 \times 10^5$$

CHECKPOINT

1 PTS

$$K_{p2} = \frac{z^2}{x_1 y_1} = 3.63 \times 10^5$$

$$K_{p4} = \frac{K_{p2} K_{p3}^2}{K_{p1}} = 3.13 \times 10^{-6}$$

CHECKPOINT

1 PTS

$$K_{p4} = \frac{K_{p2} K_{p3}^2}{K_{p1}} = 3.13 \times 10^{-6}$$

4 Incorrect.

3. In this system, reaction 1+ reaction 4 − 2× reaction 3 = reaction 2, there are 3 independent reactions.

CHECKPOINT

1 PTS

There are 3 independent reactions.

3 Correct.

Therefore, the correct option is I.